

1. A Cobb-Douglas production function relates production (Q) to factors of production - capital (K), labor (L), and raw material (M) - and an error term u using the equation

$$Q = \lambda K^{\beta_1} L^{\beta_2} M^{\beta_3} e^u, \quad (1)$$

where $\lambda, \beta_1, \beta_2$, and β_3 are the production parameters. In addition, $e^u = \exp(u)$. Suppose you have data on production and the factors of production from a random sample of firms with the same Cobb-Douglas production function. How would you use regression to estimate the production parameters? Give the specific interpretation of the parameters β_1 and β_2 in your model.

2. A study compares total compensation among top executives in a large set of US public corporations. The linear regression result is as follows:

$$\begin{aligned} \widehat{\log(Earnings)} = & 3.86 - 0.28Female + 0.37\log(MarketValue) + 0.004Return, \quad (2) \\ & (0.03) \quad (0.04) \quad (0.004) \quad (0.003) \\ n = & 46,670, \quad R^2 = 0.345, \end{aligned}$$

where *Female* is a dummy variable that is equal to 1 for females and 0 for males. *MarketValue* denotes the market value of the firm (a measure of firm size, in millions of dollars), and *Return* denotes the stock return (a measure of firm performance).

- (a) The estimated coefficient on *Female* is -0.28. Explain what this value means.
 - (b) The estimated coefficient on $\log(MarketValue)$ is 0.37. Explain what this value means.
3. With regard to the analysis of test scores and class size, an educator comments, “In my experience, student performance depends on class size, but not in the way your linear regressions say. Rather, students do well when class size is less than 20 students and do very poorly when class size is greater than 25. There are no gains from reducing class size below 20 students, the relationship is constant in the intermediate region between 20 and 25 students, and there is no loss to increasing class size when it is already greater than 25”. The educator is describing a “threshold effect”, in which performance is constant for class size less than 20, jumps and is constant for class sizes between 20 and 25 students, and then jumps again for class sizes greater than 25. To model these threshold effects, we can define dummy variables. For example, we can define $STR_{small} = 1$ if $STR < 20$, and $STR_{small} = 0$ otherwise, etc.
- (a) How many dummy variables should we use in this case? Explain.

- (b) Write down your regression model and give the interpretation of all the parameters (coefficients) in your model.
- (c) Sketch the population regression function relating Test Score to STR that is consistent with the educator's statement.